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UNIVERSAL CONTEXT-AWARE EDA & DATA ORGANIZATION REPORT
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Source File   : upcoming-ultra-trails.csv (Original Untouched)
Detected Context : General Tabular Analytics
Primary Metric   : distance_km
Generated On   : 2026-10-25 21:47:12

ANALYTICAL CAVEAT

  Findings are descriptive and correlational based on observed records; verify via controlled experimentation.

DATASET HEALTH & STRUCTURE AT A GLANCE

  Rows (Raw -> Clean)   : 1,277 -> 1,277
  Columns (Raw -> Clean)   : 13 -> 15
  Source Quality Score   : 92.0 / 100
  Duplicates Removed   : 0 (of 0 detected)
  Genuine Numeric Metrics : 3 (Excluded 4 ID/noise cols)
  Total IQR Outliers   : 215

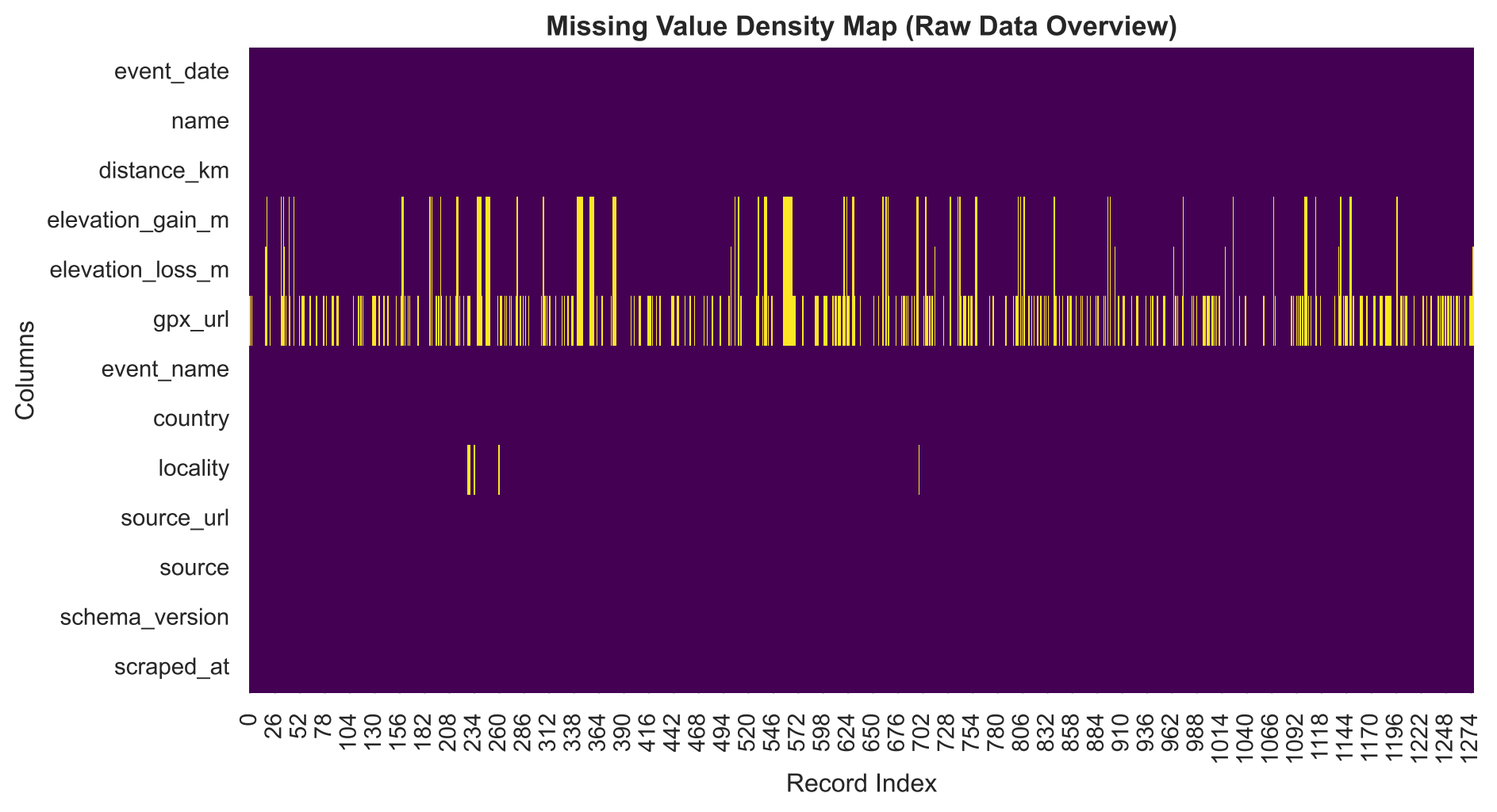
KEY CONTEXTUAL INSIGHTS
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  1. Detected Dataset Context: General Tabular Analytics.
  2. Automatically excluded 4 ID/text-noise column(s) from statistical charts.
  3. Strong linear relationship between 'elevation_gain_m' and 'elevation_loss_m' (r = 0.83).
  4. Primary Metric 'distance_km': Mean = 89.53, Median = 67.61 (Skewness = 28.60).
  5. 'Friday' leads in 'distance_km' (mean: 116.28 across 185 records).
  6. Outlier Hotspot: 'elevation_gain_m' contains 81 IQR outliers (6.3% of rows).

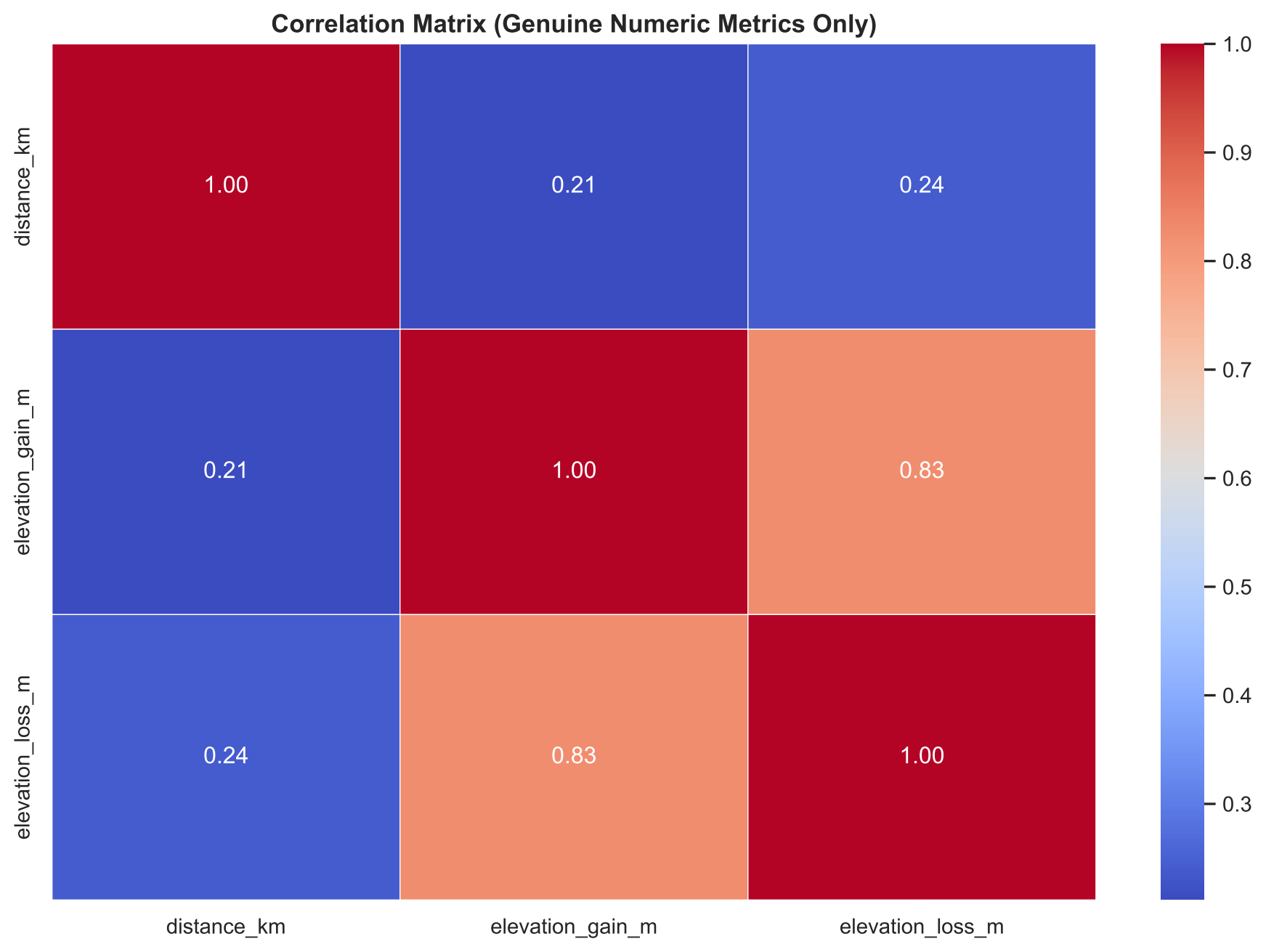
PRESCRIPTIVE RECOMMENDATIONS FOR STRATEGY & ACTIONS
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Tailored Domain Context: General Tabular Analytics
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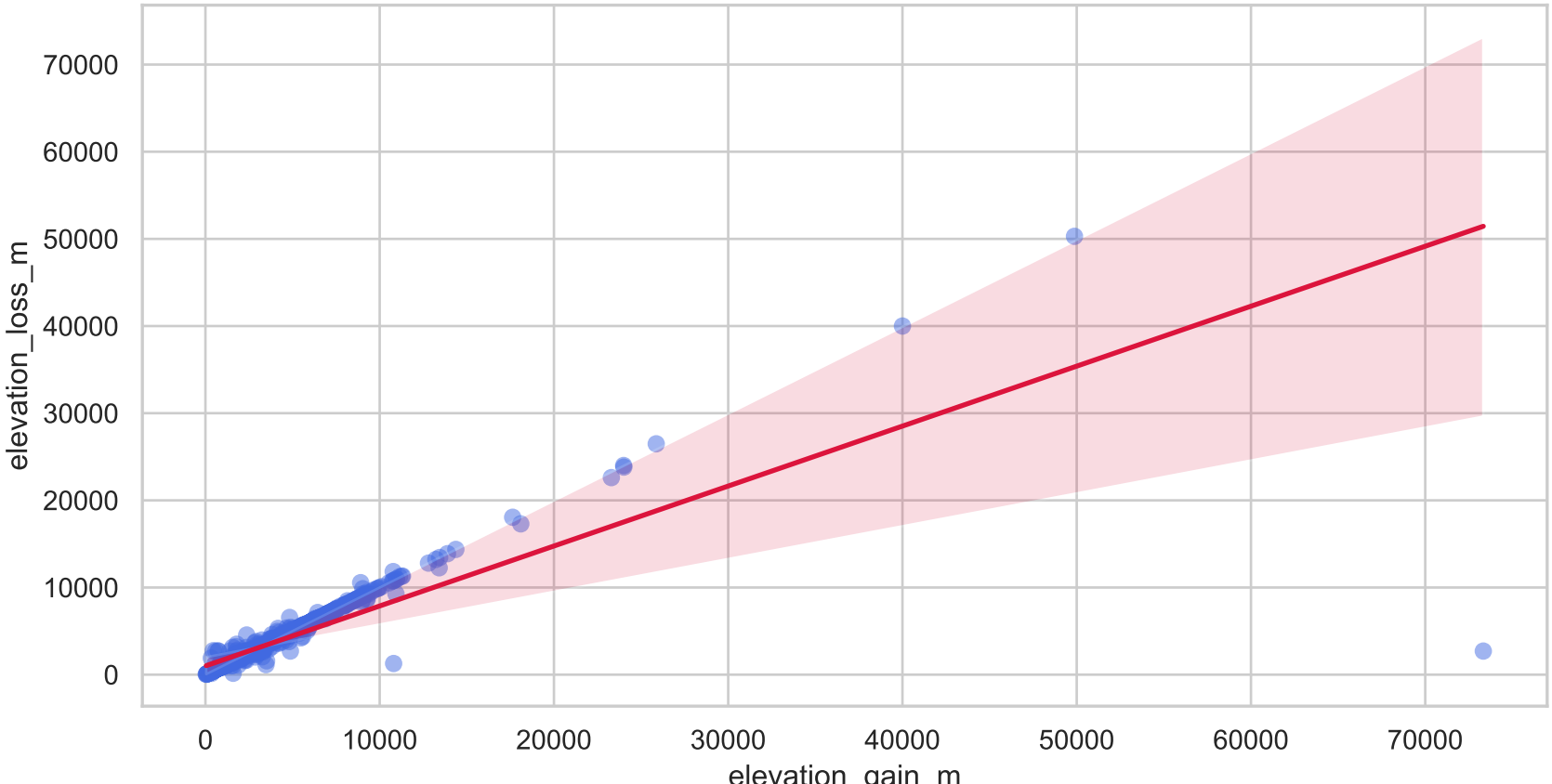
1. Address columns exceeding 50% missingness before building predictive models.
2. Transform highly skewed numeric features using log or Box-Cox scaling.
3. Cross-reference IQR outlier records with domain experts prior to removal.

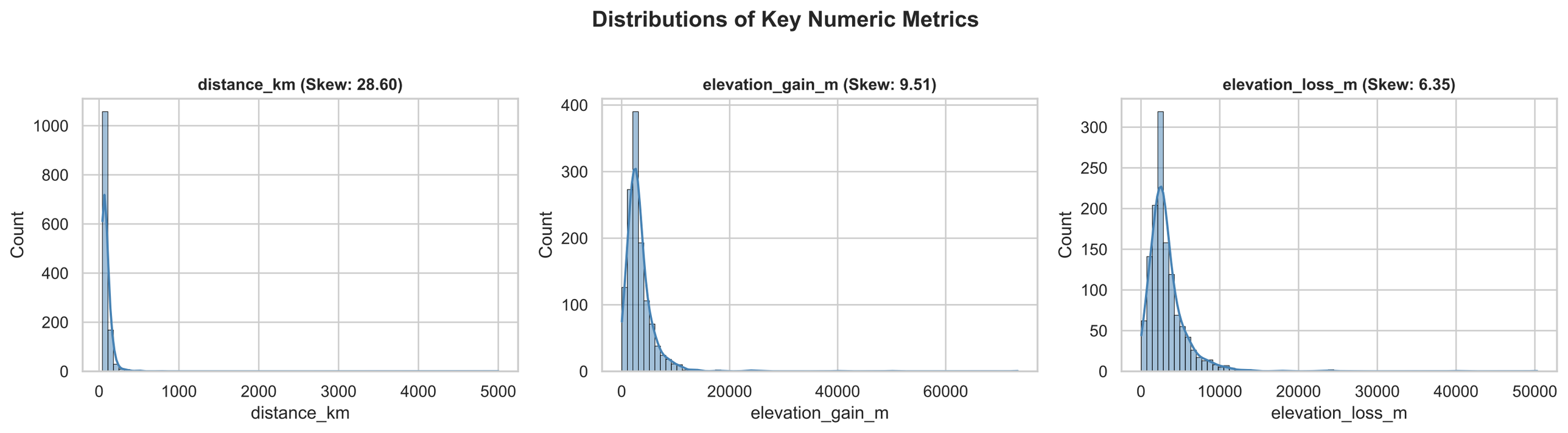
CLEANING AUDIT LOG & DATA DICTIONARY

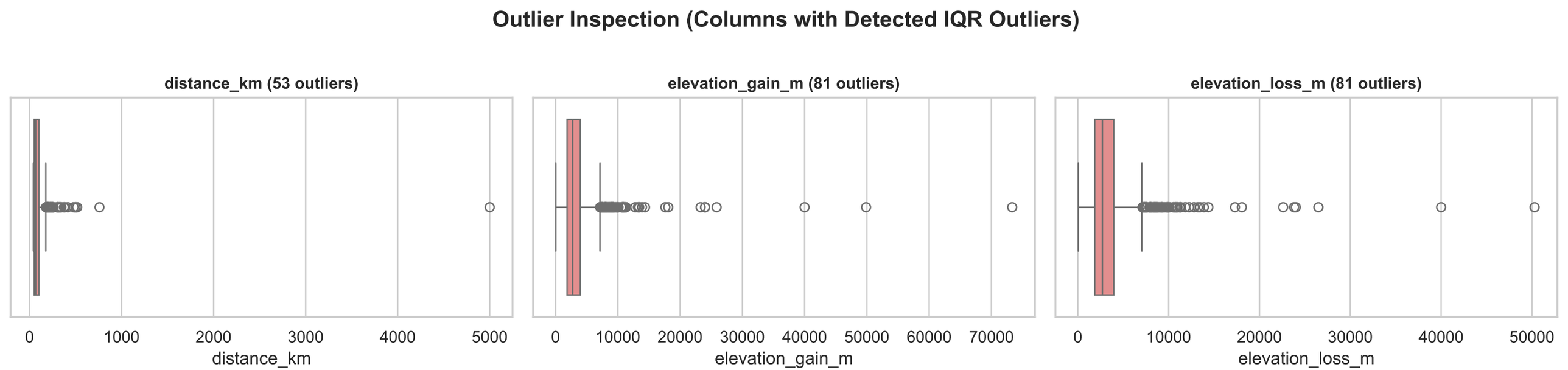
COLUMN NAME	DTYPE	NON-NULL	UNIQUE	ASSIGNED ROLE
event_date	datetime64[us	1,277	186	Date / Aux
name	str	1,277	1,016	ID / Text (Ignored)
gpx_url	str	1,277	868	ID / Text (Ignored)
event_name	str	1,277	710	ID / Text (Ignored)
country	str	1,277	84	Date / Aux
locality	str	1,277	689	Date / Aux
source_url	str	1,277	1,277	ID / Text (Ignored)
source	str	1,277	1	Date / Aux
scraped_at	str	1,277	1	Date / Aux
distance_km	float64	1,277	931	Primary Metric
elevation_gain_m	float64	1,277	1,013	Numeric Metric
elevation_loss_m	float64	1,277	995	Numeric Metric
schema_version	int64	1,277	1	Date / Aux
event_date_month	str	1,277	14	Date / Aux
event_date_dayofweek	str	1,277	7	Category Group

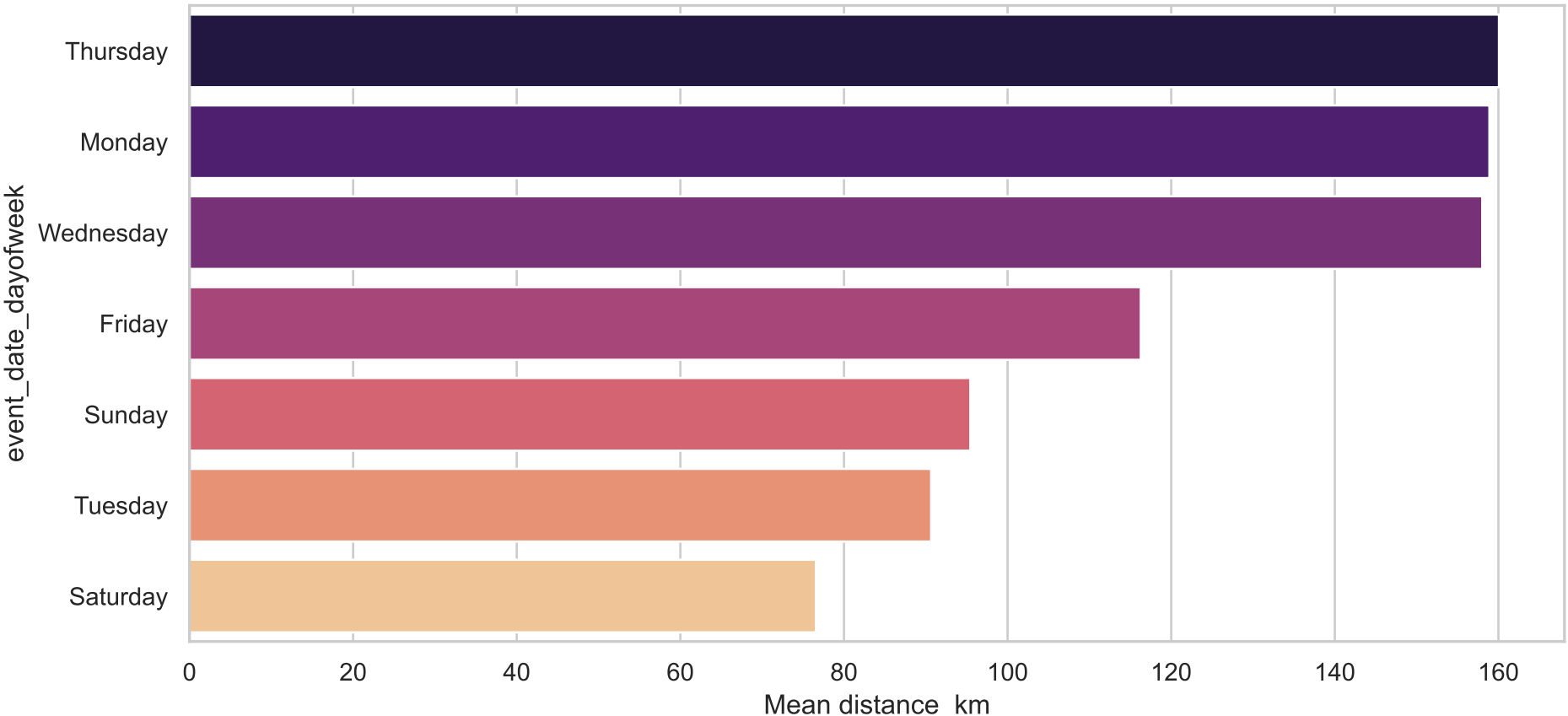

Missing Value Density Map (Raw Data Overview)
Columns
event_date
name
distance_km
elevation_gain_m
elevation_loss_m
gpx_url
event_name
country
locality
source_url
source
schema_version
scraped_at
0
26
52
78
104
130
156
182
208
234
260
286
312
338
364
390
416
442
468
494
520
546
572
598
624
650
676
702
728
754
780
806
832
858
884
910
936
962
988
1014
1040
1066
1092
1118
1144
1170
1196
1222
1248
1274
Record Index

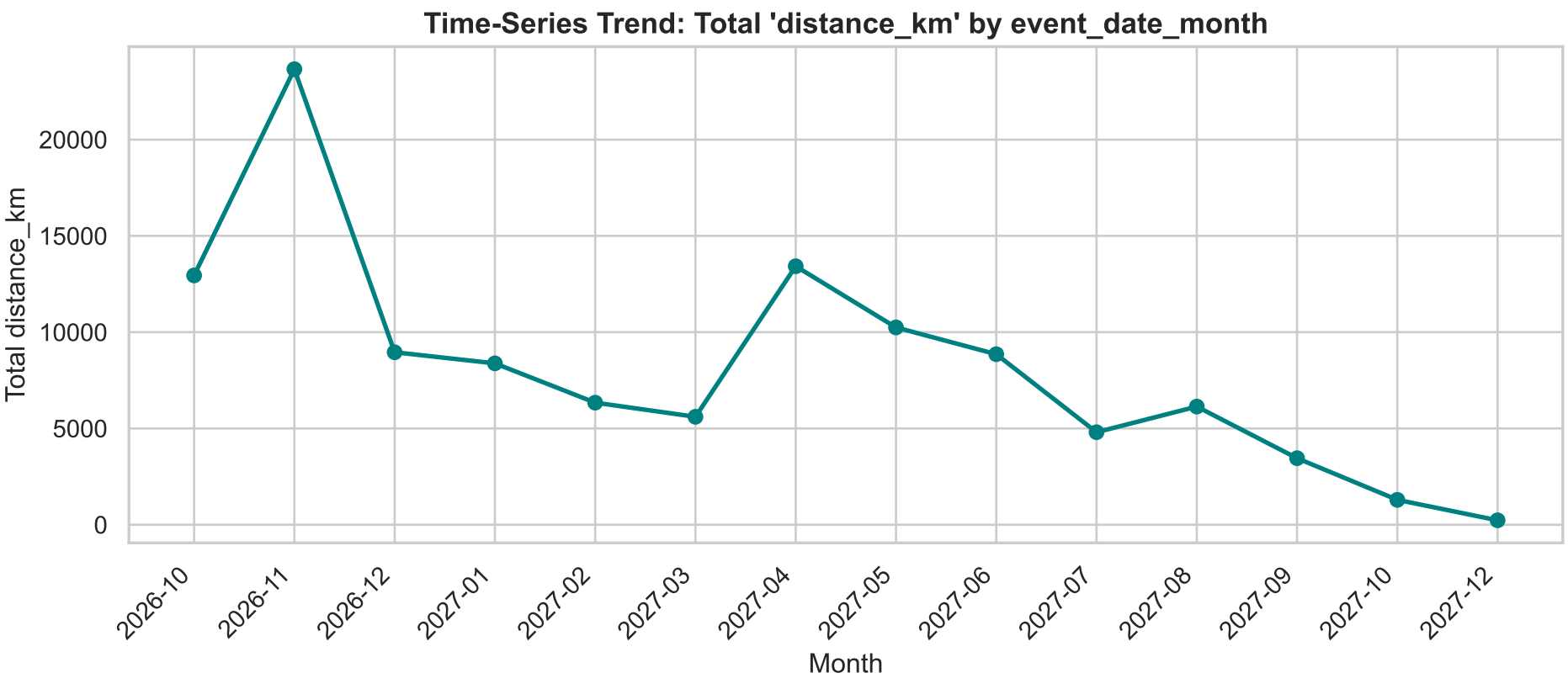

Correlation Matrix (Genuine Numeric Metrics Only)
distance_km
elevation_gain_m
elevation_loss_m
1.00
0.21
0.24
0.21
1.00
0.83
0.24
0.83
1.00
distance_km
elevation_gain_m
elevation_loss_m
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3


Strongest Relationship: 'elevation_gain_m' vs 'elevation_loss_m' (r = 0.83)
elevation_loss_m
elevation_gain_m
0
10000
20000
30000
40000
50000
60000
70000


Distributions of Key Numeric Metrics
distance_km (Skew: 28.60)
Count
distance_km
0
200
400
600
800
1000
0
1000
2000
3000
4000
5000
elevation_gain_m (Skew: 9.51)
Count
elevation_gain_m
0
100
200
300
400
0
20000
40000
60000
elevation_loss_m (Skew: 6.35)
Count
elevation_loss_m
0
50
100
150
200
250
300
0
10000
20000
30000
40000
50000


Outlier Inspection (Columns with Detected IQR Outliers)
distance_km (53 outliers)
0
1000
2000
3000
4000
5000
distance_km
elevation_gain_m (81 outliers)
0
10000
20000
30000
40000
50000
60000
70000
elevation_gain_m
elevation_loss_m (81 outliers)
0
10000
20000
30000
40000
50000
elevation_loss_m


Average 'distance_km' across 'event_date_dayofweek'
Thursday
Monday
Wednesday
Friday
Sunday
Tuesday
Saturday
event_date_dayofweek
0
20
40
60
80
100
120
140
160
Mean distance_km


Time-Series Trend: Total 'distance_km' by event_date_month
Total distance_km
0
5000
10000
15000
20000
2026-10
2026-11
2026-12
2027-01
2027-02
2027-03
2027-04
2027-05
2027-06
2027-07
2027-08
2027-09
2027-10
2027-12
Month